

CODE-

1. Write a program for process creation using C

- **Orphan Process**

```
root@LAPTOP-TUALCH80: ~
GNU nano 6.2 orphan.c
#include <stdio.h>
#include <unistd.h>

int main() {
    if (fork() == 0) {
        sleep(5);
        printf("Child Process\n");
        printf("PID = %d\n", getpid());
        printf("PPID = %d\n", getppid());
    }
    else {
        printf("Parent Process exited\n");
    }
    return 0;
}
```

- **Zombie Process**

```
root@LAPTOP-TUALCH80: ~
GNU nano 6.2 zombie.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    if (fork() == 0) {
        printf("Child Process exited\n");
    }
    else {
        sleep(10);
        system("ps -l");
    }
    return 0;
}
```

2. Create the process using fork 0 system cgl' '

- **Child Process creation**
- **Parent Process creation**
- **PPID and PID**

root@LAPTOP-TUALCH8Q: ~

fork.c

```
#include <stdio.h>
#include <unistd.h>

int main() {
    pid_t pid;

    pid = fork(); // create process

    if(pid == 0) {
        // Child process
        printf("Child Process\n");
        printf("PID = %d\n", getpid());
        printf("PPID = %d\n", getppid());
    }
    else {
        // Parent process
        printf("Parent Process\n");
        printf("PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);
    }

    return 0;
}
```